

Additional chapters appear as ellipses. They bring two kinds of information. On the one hand, theoretical background, such as precision and trueness parameter estimation and how to compute them, may be useful to better understand statistical developments involved in the method accuracy profile. On the other hand, specific examples of MU applications. One is devoted to the limits of quantification and the challenging question of controlling samples with low analyte concentration, another to method comparison.

Several data sets provide the link between the different chapters. They are used throughout for practical data handling and real software application. The aim of this data-oriented presentation is to help the analyst apply the proposed techniques in the laboratory, in keeping with the title “Companion.” This also practicality means that numerical applications for all topics covered are presented and illustrated alongside the theoretical considerations. These are based on detailed Microsoft Excel® worksheets or free equivalent, such as OpenOffice® Calc, included with the book. This software is user-friendly and does not require much explanation, and probably everyone in the laboratory knows how to use it. Although criticized by professional statisticians (for good reasons), this software is extremely helpful for quick and simple statistical computation in a laboratory, and several pitfalls can easily be avoided:

- Worksheet cell content is easily modified without any warning. Thus, once created and validated, the best initiative is to protect the worksheet or whole workbook.
- The formula inside cell is not visible unless the option to show formulas is on. To help the understanding of the template worksheets developed for this book, all formulas are made visible in the cell next to the resulting. The built-in function `FORMULATEXT` is used for this aim. It is only available in the most recent Excel releases.
- Confusion may exist between a worksheet and a text editor. Fancy presentation must be avoided, and it is better to embed a worksheet within a text editor rather than trying to do everything with a single software.

The basic use of worksheet software does not allow complex statistical calculation though it contains many built-in functions, which are used in the following examples. It is possible to use the development environment called Visual Basic for Applications coming with Excel to build more complex programs, but it requires some practice. For the most sophisticated applications, we preferred to provide Python program examples. This software is increasingly popular, and the accuracy of statistical functions is widely recognized. For instance, complex techniques, such as non-linear or weighted regression techniques, are easily implemented. Python is simpler than professional statistical software. It is developed under a free license, and there is an exceptionally large community of users who can help. The drawback is that it is a patchwork, and many additional modules must be imported to apply some methods. The simplest way to install Python is to download a free package called Anaconda [14] and select the Spyder development environment. Presented examples were programmed in this environment.

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Glossary of Symbols

Symbol	Term
β	Coverage probability of the tolerance interval
$u(Z)$	Standard uncertainty Z
r	Coefficient of correlation
Z	Inverse-predicted concentration in the working sample
Z^*	Extrapolated sample concentration (standard addition method)
Y	Measured instrumental response
$\hat{Y}$	Predicted instrumental response
X	Concentration of the (authentic) analyte in the working sample
$\bar{X}, \bar{Z}$	Average
$\bar{\bar{X}}, \bar{\bar{Z}}$	Grand average
X_c	Concentration of the (surrogate or not) analyte in the calibrant
UR%	Relative expanded standard uncertainty
$U(Z)$	Expanded uncertainty Z
AIC	Akaike Information Coefficient
A	Variance ratio $A = s_B^2/s_r^2$
δ	Bias
E	Random error variable
f	Any calibration or uncertainty function
f^{-1}	Inverse of any function
β - γ -CTI	β - γ -Content Tolerance interval
β -ETI	β -Expectation Tolerance interval
CF	Correction factor
$p\%$	Proportional correction factor
AA	Authentic analyte (used as subscript)
IS	Internal Standard (used as subscript)

Symbol	Term
$1 - \alpha$	Level of confidence (also noted γ)
$[A-, A+]$	Acceptance interval
$u_c(Z)$	Combined standard uncertainty Z
$u^2(Z)$	Standard variance of Z
RF	Response Factor
SP	Sum of crossed products of deviations to the mean
SS	Sum of squared deviations to the mean
s_r^2	Repeatability variance
s_W^2	Within-series variance
s_R^2	Reproducibility variance
s_L^2	Between-laboratories variance
s_{IP}^2	Intermediate precision variance
s_B^2	Between-series variance
r^2	Coefficient of determination
k_{TI}	Tolerance factor of a tolerance interval
k_{GUM}	Coverage factor
k_{GUM}	Standardized coverage factor (GUM)
$a_0, a_1, a_2, \dots$	Coefficients of the calibration model
$[Z \pm U(Z)]$	Coverage interval
G_n	Input quantity of the measurement model
$[X_L, X_U]$	Measuring interval or working interval

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